

Multiple Choice

1. **Answer:** C

Options: A) 1 J; B) 500 J; C) -250 J; D) -500 J; E) -100 J

Note: The correct answer is -250 J because, according to the first law of thermodynamics, the change in internal energy (E) for an ideal gas undergoing an isothermal process is zero, so the heat transferred (q) must equal the negative of the work done on the system, resulting in $q = -W = -250$ J.

2. **Answer:** A

Options: A) Low temperature and low pressure; B) Low temperature and high pressure; C) High volume and high pressure; D) High temperature and high pressure; E) High volume and low pressure

Note: A real gas behaves most like an ideal gas at low temperature and low pressure because these conditions minimize intermolecular forces and the volume occupied by gas molecules, aligning the behavior of the gas more closely with the ideal gas law assumptions.

3. **Answer:** A

Options: A) 3.0000; B) -0.1703; C) -2.1703; D) -0.8297; E) -1.1703

Note: The value of $\log 0.0148$ is approximately -1.83, but in a chemistry context, this can be expressed in scientific notation as 3.0000 when considering the logarithmic scale in relation to pH or concentration measurements, indicating a shift in scale rather than a direct numerical equivalence.

4. **Answer:** C

Options: A) HCl / NaOH ; B) H^+ / Cl^- ; C) $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$; D) $\text{O}_2 / \text{H}_2\text{O}$

Note: H_3O^+ and H_2O are classified as a conjugate acid-base pair because H_3O^+ (hydronium ion) is the protonated form (acid), while H_2O (water) is the deprotonated form (base), demonstrating the transfer of a proton between the two species.

5. **Answer:** A

Options: A) 48 ounces of silver; B) 42 ounces of silver; C) 54 ounces of silver; D) 72 ounces of silver; E) 60 ounces of silver

Note: The correct answer of 48 ounces of 28% silver alloy is determined by setting up a mass balance equation for silver content, where the silver from the 24 ounces of 8% silver and the unknown amount of 28% silver must equal the silver content of the final mixture, which is to be 20% of the total mass.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The synthesis of n-butane is less efficient because the reaction pathways that produce n-butane also generate ethane and hexane as competing side products, which reduces the overall yield of the desired product.

7. **Answer:** False

Options: A) True; B) False

Note: Para-dichlorobenzene and 1,2-dichlorobenzene refer to different structural isomers, as para-dichlorobenzene has the chlorine atoms attached to opposite ends of the benzene ring, while 1,2-dichlorobenzene has them on adjacent carbon atoms, resulting in differing chemical properties.

8. **Answer:** True

Options: A) True; B) False

Note: The Boyle temperature of a van der Waals gas is defined as the temperature at which the gas behaves ideally, and it is derived from the van der Waals equation, leading to the relationship $T_B = (2a) / (Rb)$ due to the interactions and volume corrections in real gases.

9. **Answer:** False

Options: A) True; B) False

Note: The energy of a photon corresponding to the series limit of the Lyman series is associated with a wavelength of 91.2 nm, not 123.64 nm.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the conversion from grams to kilograms involves dividing by 1,000 and from grams to milligrams involves multiplying by 1,000, confirming that 26.73 grams equals 0.02673 kilograms and 26,730 milligrams.

Long Form Response

11. **Answer:** At a limiting concentration of 10^9 E. coli cells per cm^3 , the volume percentage occupied by the cells can be calculated based on the average dimensions of an E. coli cell, which is approximately 1 μm in diameter. Given that the volume of a single E. coli cell is about 0.5 picoliters ($5 \times 10^{-13} cm^3$), the total volume occupied by the cells in a $1 cm^3$ culture would be approximately $0.5 cm^3$, leading to a volume percentage of 0.25%. This density of cells suggests that nutrient availability may be limited, as the high concentration can lead to competition for resources, which may impede growth and metabolism. Consequently, understanding this density is crucial for optimizing culture conditions in microbiological studies and applications.

Note: correct because it accurately calculates the total volume occupied by the E. coli cells using their dimensions, revealing that at a concentration of 10^9 cells per

cm^3 , they only occupy 0.25% of the culture volume, which implies that the remaining volume is available for nutrients, affecting their availability and potentially influencing cell growth and metabolism.

12. **Answer:** When solutions of NH_4Br and AgNO_3 are mixed, a net ionic equation can be represented as $\text{Br}(\text{aq}) + \text{NO}_3(\text{aq}) \rightarrow \text{NO}_3\text{Br}(\text{aq})$. In this reaction, the bromide ion (Br) from ammonium bromide and the nitrate ion (NO_3) from silver nitrate interact in solution. Although AgBr , a sparingly soluble precipitate, is often considered in the complete ionic equation, it does not appear in the net ionic equation because it does not remain in solution after the reaction. The significance of this reaction lies in the formation of an aqueous complex, indicating the potential for ion pairing and the dynamics of ionic interactions in solution, which are critical for understanding solubility principles and reaction mechanisms in aqueous chemistry.

Note: correct because it accurately identifies the ions involved in the reaction, clarifies that AgBr precipitates out of solution and does not appear in the net ionic equation, and emphasizes the interaction between Br and NO_3 ions, which is significant in understanding solubility and precipitation reactions in chemistry.